

# How To Improve POWER CONVERSION EFFICIENCY Of INVERTERS

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*This article compares different inverter types and provides tips for inverter designers to ensure high-quality power at higher conversion efficiencies*

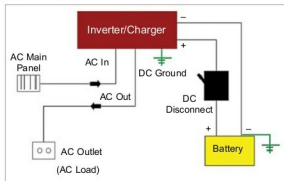
**I**nverters convert DC electricity from sources like solar panels, batteries and fuel cells into AC electricity. Their power-handling capacities like input voltage, output voltage and frequency depend on their design.

Inverters require a stable DC power source that can supply enough current for the required power demand. Apart from the design, the input voltage of an inverter depends on its intended application as follows:

1. 12V DC for smaller consumer and commercial inverters
2. 24V, 36V and 48V DC for home energy systems
3. 200-400V DC for photovoltaic solar panels
4. 300-450V DC for electric vehicle battery packs in vehicle-to-grid systems
5. Hundreds of thousands of volts for high power transmission systems

A wide range of applications including portable consumer devices, hybrid/electric vehicles, industrial control systems and solar power systems are driving the demand for inverters as these ensure a high-efficiency and high-reliability power source. Inverters help to save energy over conventional on/off control. Their faster switching enables more precise temperature control and reduced audible noise (in fan motors). Use of semiconductors in inverters helps to boost performance, minimise power losses and optimise thermal management.

*Block diagram of the conversion process in an inverter (Source: <http://engineering.electrical-equipment.org>)*



## Applications of inverters

Inverters find diverse applications due to their DC-to-AC conversion function. Uninterrupted power supply (UPS) systems utilise inverters with

a set of batteries to supply backup power when the mains power is not available. Inverter technology is advancing further, and many types of inverters are being manufactured and designed to serve different needs.

Other than for emergency backup power in homes, inverters are also used in some aircraft systems (to convert a portion of the aircraft DC power into AC), electric motor speed control, refrigeration compressors, power grid-tied solar power systems, induction heating, electric shock devices, etc.

## Types of inverters

Different types of inverters are available for different applications: multilevel inverters, voltage source inverters, sinewave inverters, modified sinewave inverters and squarewave inverters. Voltage source inverters provide the required range of output voltage or current and frequencies. As these ranges are either positive or negative, voltage source inverters are known as two-level inverters. Multilevel inverters synthesise a near sinusoidal voltage from several levels of DC voltages. Minimum harmonic distortion is their main advantage.

Most of the pure sinewave inverters have good accuracy and very high load capacity, but these are complex in design and expensive. They power devices with less power loss and less heat generation. Sinewave inverters can be used for providing backup power to battery chargers, electric drills, electric tube-lights, digital clock radios and other sensitive electronics.

The third and fifth harmonics in the output of a squarewave inverter may burn power and severely impact the efficiency of devices. These inverters are used in low-cost AC motor drives, induction heating units and electronic ballasts for fluorescent lamps.

Modified sinewave inverters can be used for most household electrical devices. However, these have a large amount of power efficiency loss due to significant harmonic